ASME PONGO

Professional Outreach & Network Geographic Overview

Mapping ASME Professional and Student Sections, jurisdictions, and geographic reach across the United States.

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User Guide
Professional + Student Section Edition

Core concept: PONGO combines several geographic layers that answer different questions. Professional Section jurisdiction represents published county/parish territory. Professional and Student modeled reach are adjustable distance-based planning assumptions. These layers should be compared, but not treated as equivalent.

1. What is ASME PONGO?

ASME PONGO - Professional Outreach & Network Geographic Overview - is an interactive geographic planning dashboard for exploring ASME Professional Sections and Student Sections across the continental United States.

PONGO is designed to support outreach planning, section collaboration, student-to-professional engagement, geographic analysis, and discussion of potential expansion opportunities.

2. What PONGO Can Show

PONGO can display Professional Section headquarters, Student Section campus locations, adjustable modeled reach around both types of sections, published Professional Section county/parish jurisdictions, modeled geographic overlap and gaps, nearest neighboring Professional Sections, and hypothetical expansion hubs.

3. Map Color and Layer Guide

Map element	Display	Meaning
Professional headquarters	Blue marker	Headquarters/city location of a Professional Section.
Professional modeled reach	Blue transparent circle	Adjustable 10-150 mile planning radius.
Student Section headquarters	Orange marker	University/campus location associated with a Student Section.
Student modeled reach	Orange transparent circle	Independent adjustable 10-150 mile planning radius.
Official Professional jurisdiction	Green polygon	County/parish territory mapped from ASME Professional Section descriptions.
Partial jurisdiction	Hatched green polygon	ASME description covers only part of the county; exact sub-county geometry is not shown

4. Professional Sections Search

The **Professional Sections** search is dedicated only to Professional Sections. Choose a section and add it to the active selection. Multiple Professional Sections can be selected simultaneously.

Clear selected removes Professional Section selections. **Select all** explicitly selects all Professional Sections in the dashboard. Professional selections operate independently from Student Section selections.

5. Student Sections Search

The separate **Student Sections** search allows universities and campus-based ASME Student Sections to be selected independently from Professional Sections.

Multiple Student Sections can be displayed simultaneously. **Clear students** removes Student Section selections, while **Select all students** displays all Student Sections contained in the current PONGO dataset.

6. Using Professional and Student Searches Together

The two search controls can be used at the same time. For example, a user can select one Professional Section and several nearby Student Sections to examine possible collaboration, outreach, mentoring, transition, or joint-programming opportunities.

Because the two selections are independent, changing the Student Section list does not remove selected Professional Sections, and vice versa.

7. State and Territory Filters

The **State** filter restricts displayed locations by state. The **Territory** filter provides broader Professional Section regional groupings such as Northeast, Midwest, Southeast, Southwest, and Northwest.

Filters can be combined with the Professional and Student multi-select controls to narrow the analysis.

8. Professional Modeled Radius

The Professional Section Coverage Radius slider ranges from **10 to 150 miles** in 5-mile increments. It controls the blue modeled-reach circles around Professional Section headquarters.

This radius is an analytical assumption. It does not modify or redefine official ASME county/parish jurisdiction.

9. Student Section Modeled Radius

Student Sections have their own independent **10 to 150 mile** radius control. The resulting modeled reach is shown in orange.

This makes it possible to model Professional and Student reach differently. For example, Professional Sections can be modeled at 75 miles while Student Sections are modeled at 25 miles.

10. Radius Interpretation

Radius	Possible planning interpretation
10 mi	Immediate campus/local area
25 mi	Local metropolitan engagement
50 mi	Regional commuting/engagement area
75 mi	Broader regional reach
100 mi	Multi-city reach
150 mi	Large regional planning area

These interpretations are examples for planning and are not official ASME service-radius definitions.

11. Official ASME Jurisdiction

The **Official ASME Jurisdiction** toggle displays green county/parish polygons associated with Professional Sections. This layer is separate from both modeled-radius systems.

A jurisdiction polygon represents a geographic unit included in the Professional Section territory information used to build PONGO. Student Sections do not receive official county jurisdiction polygons because they are campus-based organizations.

12. Partial-County Jurisdictions

Some Professional Section territory descriptions cover only part of a county. PONGO uses a hatched green appearance to flag these cases.

The entire county polygon may be visible for geographic reference, but the hatching explicitly indicates that the full county should not be interpreted as the exact official territory.

13. Map Visibility Controls

Users can independently show or hide Professional modeled coverage, Official ASME Jurisdiction, Professional headquarters, Student headquarters, and Student modeled coverage.

This allows clean presentation views. For example, a user can hide all radius circles and display only Professional jurisdictions with Professional and Student headquarters.

14. Including Students in Coverage Metrics

By default, Student Section locations can be displayed without changing the Professional-oriented modeled coverage calculations.

Enable **Include students in modeled coverage metrics** when the analytical question requires both Professional and Student modeled reach to contribute to overall coverage, overlap, and gap calculations.

15. Analysis Modes

Coverage Circles: Standard view for comparing Professional and Student modeled reach.

Overlap Heatmap: Highlights sampled locations reached by multiple modeled coverage areas.

Coverage Gaps: Highlights sampled continental U.S. land outside the modeled reach of the locations currently contributing to the analysis.

Coverage gaps and heatmaps are analytical results. They are not official ASME jurisdiction determinations.

16. Professional Section Details

Click a blue Professional Section marker to inspect its name, headquarters location, regional territory, selected Professional radius, official-jurisdiction mapping status, nearest Professional Section, and neighboring sections.

Nearest-section distances are straight-line geographic distances rather than driving distances.

17. Student Section Details

Click an orange Student Section marker to open the Student Section Details panel. PONGO displays the university/section name, campus city and state, Student modeled radius, nearest Professional Section, and straight-line distance to that Professional Section.

This feature is particularly useful for exploring potential **Student Section -> Professional Section** engagement pathways.

18. Show Only This Professional Section

After selecting a Professional Section marker, **Show Only This Section** isolates it from the broader Professional Section network. This is useful for inspecting its official jurisdiction and nearby Student Sections without competing Professional polygons.

Use **Reset Filters** to return to the broader dashboard view.

19. Expansion Simulator

Select **Place Simulated Hub** and click a location on the continental U.S. map to test a hypothetical new engagement location.

The simulator reports its approximate coordinates, nearest existing Professional Section, distance to that section, and estimated new sampled modeled coverage. A simulated hub is a planning scenario only and does not imply creation of a new ASME Section or jurisdiction.

20. Dashboard Metrics

Visible Professional: Number of Professional Sections currently displayed.

Visible Student: Number of Student Sections currently displayed.

Estimated Land Covered: Approximate sampled land reached under the active modeled-radius assumptions.

Max Overlap: Highest number of modeled reach areas intersecting at a sampled location.

Average Nearest Section: Average straight-line distance between displayed Professional Section headquarters and their nearest displayed Professional neighbor.

Isolated Beyond Radius: Number of displayed Professional Sections whose nearest displayed Professional neighbor is beyond the selected Professional radius.

Simulated Hub: Indicates whether a hypothetical hub is active.

21. Export CSV

Export CSV downloads the currently visible section information and distinguishes records as **Professional Section** or **Student Section**.

Professional records include jurisdiction-mapping information and the Professional modeled radius. Student records include the Student modeled radius. The export can be used in Excel, Power BI, Tableau, Python, or other analysis tools.

22. Export Map PNG

Export Map PNG saves the current map as a high-resolution image. The exported image preserves visible Professional markers and blue radii, Student markers and orange radii, jurisdiction polygons, partial-county indicators, analysis overlays, and the simulated hub.

This is useful for presentations, reports, section-planning documents, and leadership discussions.

23. Shareable Settings URL

Copy Settings URL preserves the dashboard configuration when PONGO is hosted on a website. The saved configuration can include selected Professional Sections, selected Student Sections, Professional and Student radii, filters, layer visibility, analysis mode, and simulator settings.

This lets another user open essentially the same analytical view without manually rebuilding it.

24. Recommended PONGO Workflows

Professional Section territory review

Select a Professional Section, enable Official ASME Jurisdiction, and hide modeled coverage if you want to focus only on the published county/parish structure.

Professional vs. Student outreach

Select one or more Professional Sections and nearby Student Sections. Compare headquarters/campus locations, modeled reach, and Student-to-Professional distances.

Student pipeline analysis

Display Student Sections in a region and inspect which Professional Section headquarters are geographically closest. Use this as a starting point for evaluating mentoring, transition, and collaboration opportunities.

Modeled gap analysis

Display the desired network, select Coverage Gaps, and test several radii. Then enable Professional jurisdiction to determine whether a distance-based gap lies inside an existing section territory.

Regional collaboration

Select multiple neighboring Professional Sections and Student Sections to identify geographic clusters that may support joint events or coordinated outreach.

Expansion scenario

Use jurisdiction and gap layers to identify an area of interest, place a simulated hub, and compare its modeled reach with existing Professional and Student locations.

Presentation preparation

Configure only the layers needed for the intended message, then export the map as PNG and the supporting records as CSV.

25. How to Interpret PONGO Correctly

PONGO is strongest when its layers are used together but interpreted separately. **Jurisdiction** answers where a Professional Section's published territory lies. **Modeled reach** asks what geography may be accessible under a chosen distance assumption. **Student Section locations** show campus-based ASME presence. **Gap and overlap analysis** identifies areas worth further investigation.

A geographic gap does not automatically mean that a new Professional Section is needed, and a large modeled radius does not mean that a section officially serves every person inside that circle.

26. Important Limitations

PONGO is a geographic planning and exploratory analysis tool, not an authoritative governance record. ASME source materials remain authoritative for current Section status and territory.

Campus coordinates represent university/campus locations for geographic planning rather than precise meeting-room or building coordinates. Partial-county Professional jurisdictions are flagged because whole-county geometry cannot represent the exact sub-county boundary.

The current tool does not by itself measure actual member density, participation, travel time, section activity, population, local industry concentration, or virtual engagement.

27. Data Quality and Updates

Professional and Student Section structures can change. Before using PONGO output for formal organizational decisions, verify current Section status and jurisdiction information against current ASME records.

Future versions can improve PONGO by incorporating refreshed section directories, exact partial-county definitions where available, member-density information, Student Section activity data, and additional engagement indicators.

28. Recommended Decision Framework

1. Start with official Professional jurisdiction.
2. Add Professional headquarters and modeled reach.
3. Add Student Sections and their modeled reach.
4. Examine overlap, gaps, nearest relationships, and potential hubs.
5. Validate promising findings with membership, participation, local leadership, and current ASME organizational data before acting.

ASME PONGO at a Glance

Professional Outreach & Network Geographic Overview

BLUE = Professional Sections and modeled reach
ORANGE = Student Sections and modeled reach
GREEN = Professional Section county/parish jurisdiction

Tagline: Mapping ASME Professional and Student Sections, jurisdictions, and geographic reach across the United States.

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